

Rethinking Generative AI Applications

COMM116 Generative AI Applications

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Let's draw everything on the white board

- The spectrum from theory to application
- What would be the dimensions?

Sections

- Section 1 – Applications in Images and Videos
- Section 2 – Language, Data and Knowledge
- Section 3 – Addressing the Ethical Issues of LLMs
- Section 4 – World Models and Autonomous Systems
- Section 5 – Domain-specific Applications

Section 1 – Applications in Images and Videos

- **Generative AI** builds a model of a data distribution and then samples from it.
- Data: images, text, audio, videos, molecules, code . . .
- Goal: generate new **samples** that “look like” the training data, while being novel (not memorized; not exactly the same with the training data).
- Models: VAE, Diffusion, U-Net, Vision Transformer, Diffusion Transformer

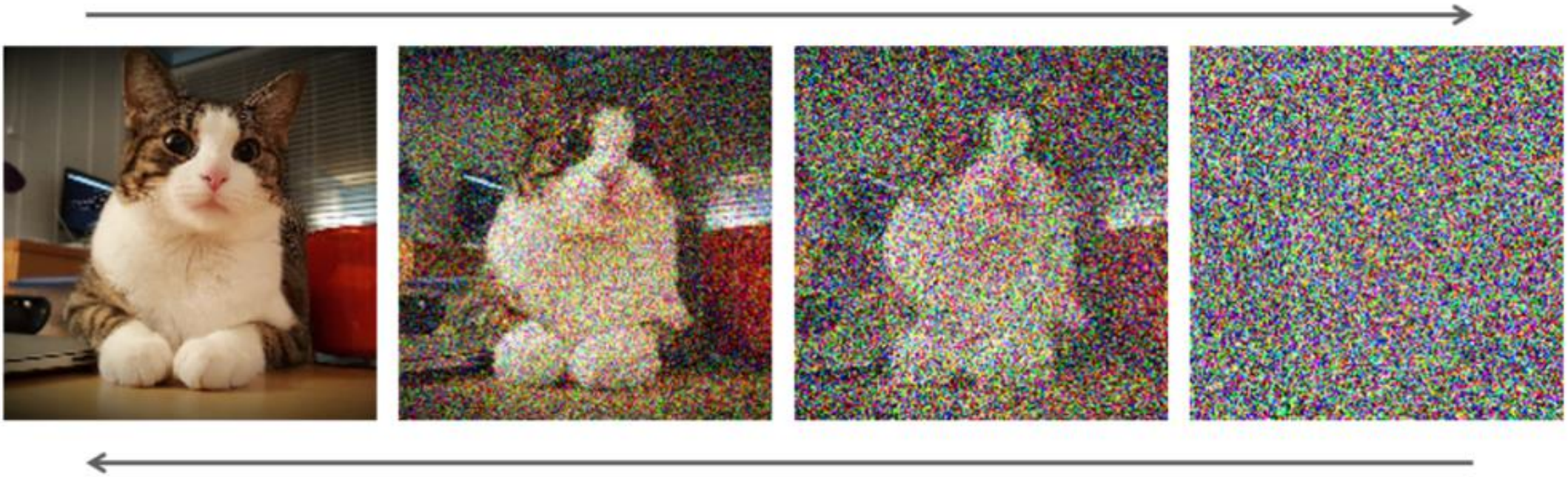


Image from Arash Vahdat and Karsten Kreis. Improving Diffusion Models as an Alternative To GANs, Part 1. Apr 26, 2022. <https://developer.nvidia.com/blog/improving-diffusion-models-as-an-alternative-to-gans-part-1/>

Section 2 – Language, Data and Knowledge

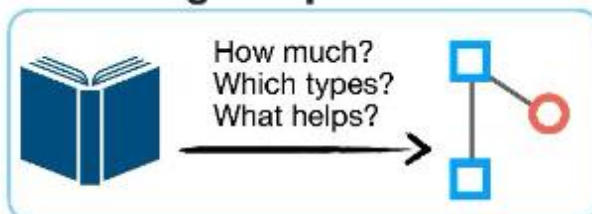
- Language Models and their limitations
- The lifecycle of knowledge in LLMs
- Potential ways to integrate knowledge into LLMs

Three Pillars of “Machine Knowledge”

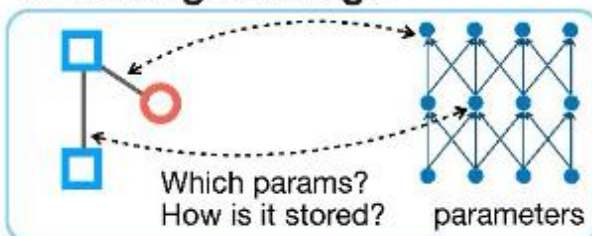
			
Representation:	crisp	mixed	<i>latent</i>
Correctness:	high	mixed	mixed
Freshness:	mostly	fresh	stale
Coverage:	limited	everything	huge
Cost/Query:	low	low	high
Provenance:	tangible	tangible	intangible
			
Coolness:	nerdy	mundane	amazing

Tutorial Roadmap

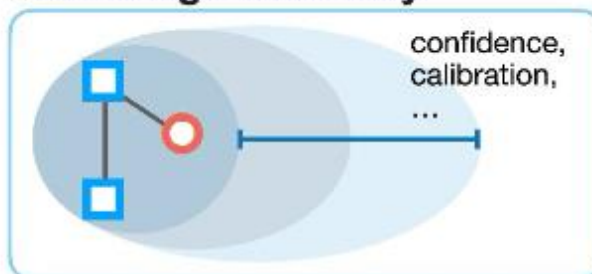
knowledge acquisition



knowledge storage

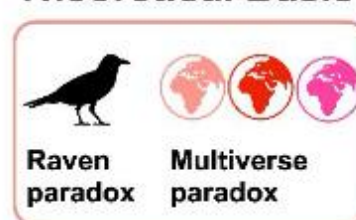


knowledge boundary

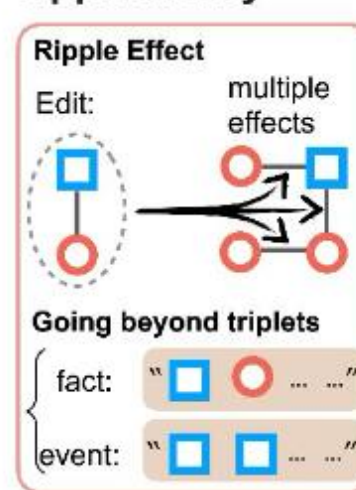


(b) Foundation: understanding LLMs' knowledge

Lack of A Theoretical Basis

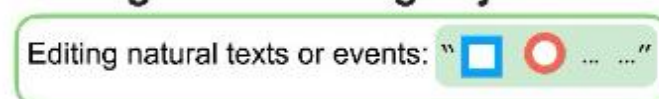


Limited Applicability

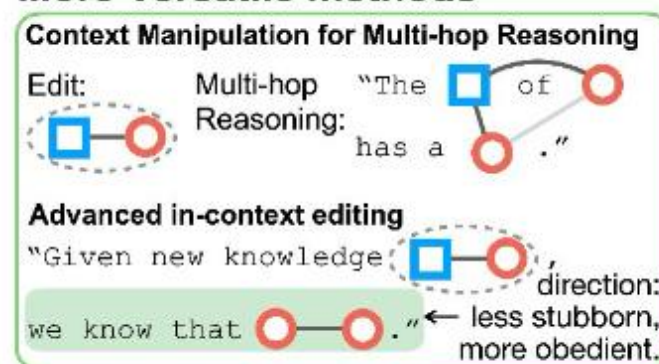


(c) Challenges in editing knowledge

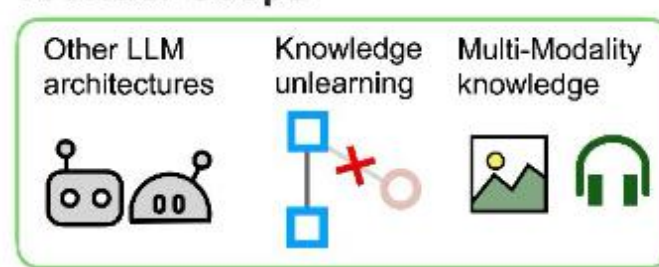
More general editing objective



More Versatile methods

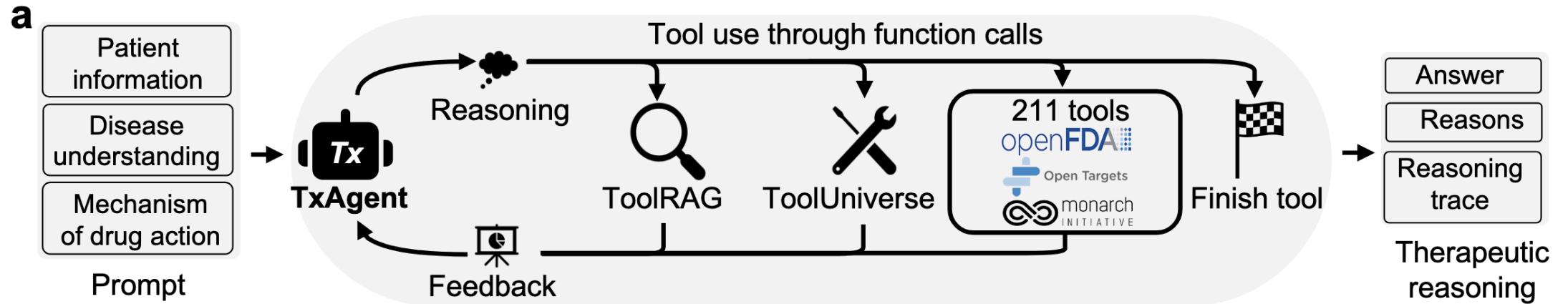


A Wider Scope



(d) Promising directions

Knowledge graphs as tools in agentic AI

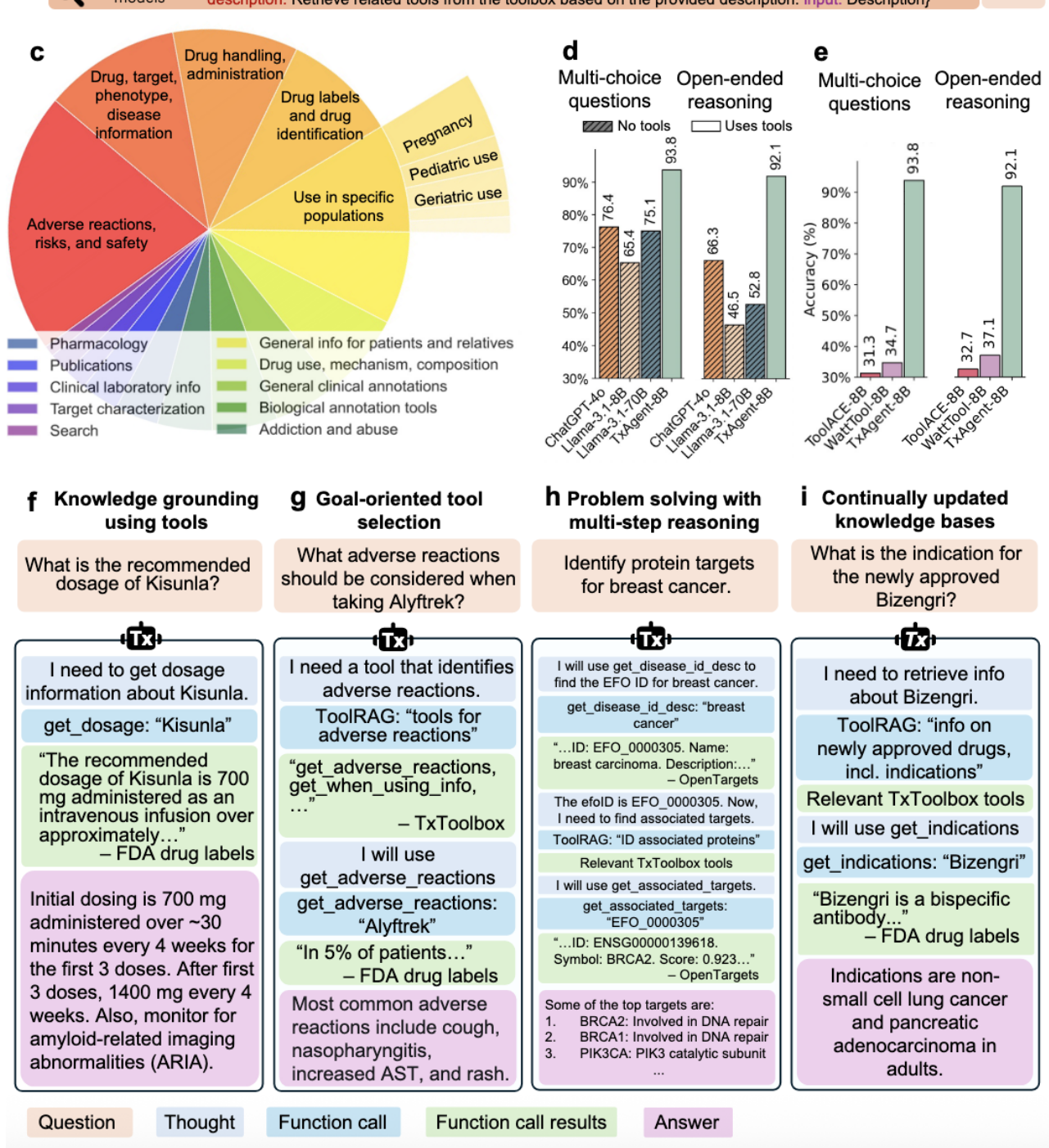


b

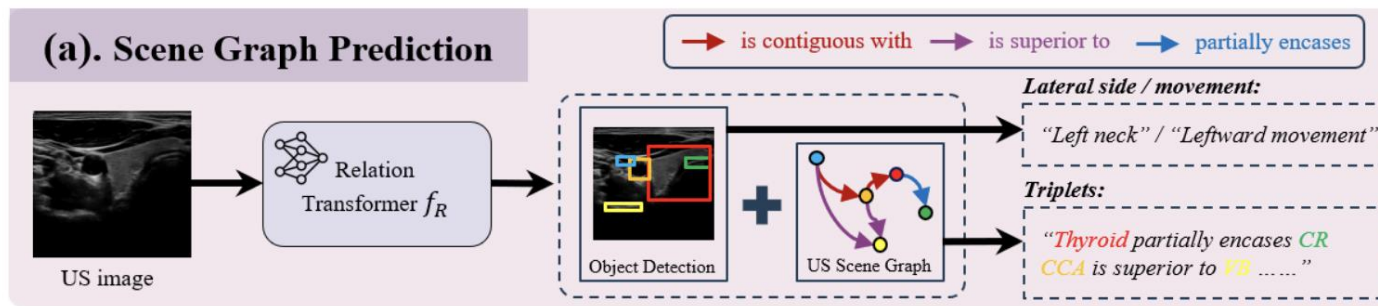
	{name: get_associated_targets_by_disease_efoID description: Find targets associated with a specific disease or phenotype based on EFO ID. input: EFO ID}	52 tools
	{name: get_contraindications_by_drug_name description: Retrieve contraindications information based on the drug name. input: Drug name}	156 tools
	{name: get_associated_diseases_by_HPO_ID description: Retrieve diseases associated with a list of phenotypes by the phenotype HPO IDs. input: HPO ID}	3 tools
	{name: Tool_RAG description: Retrieve related tools from the toolbox based on the provided description. input: Description}	1 tool

S. Gao *et al.*, 'TxAgent: An AI Agent for Therapeutic Reasoning Across a Universe of Tools', Mar. 14, 2025, *arXiv*: arXiv:2503.10970. doi: [10.48550/arXiv.2503.10970](https://doi.org/10.48550/arXiv.2503.10970).
<https://zitniklab.hms.harvard.edu/TxAgent/>

S. Gao *et al.*, ‘TxAgent: An AI Agent for Therapeutic Reasoning Across a Universe of Tools’, Mar. 14, 2025, *arXiv*: arXiv:2503.10970. doi: [10.48550/arXiv.2503.10970](https://doi.org/10.48550/arXiv.2503.10970).

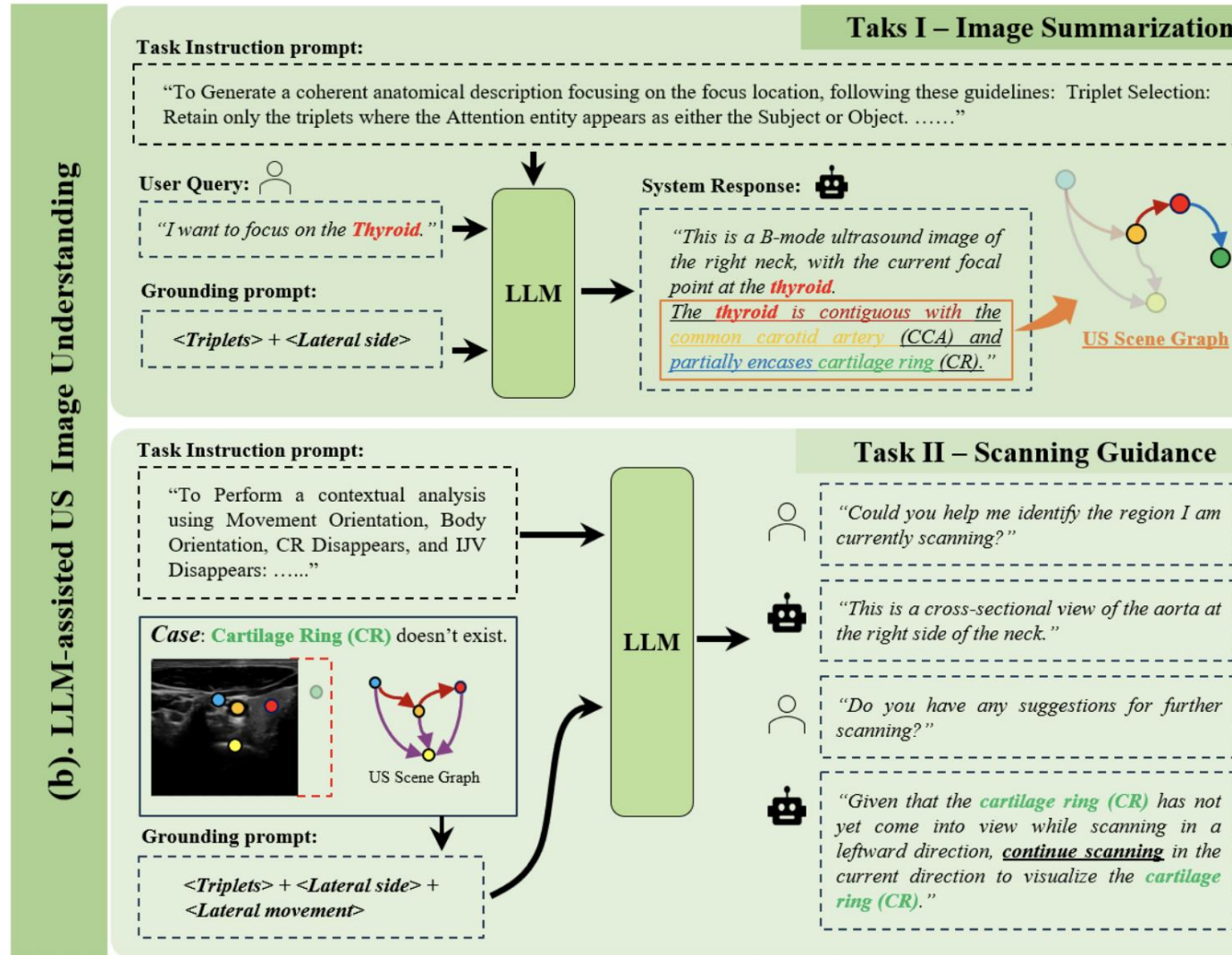


Scene graphs



Semantic Scene Graph for Ultrasound Image Explanation and Scanning Guidance

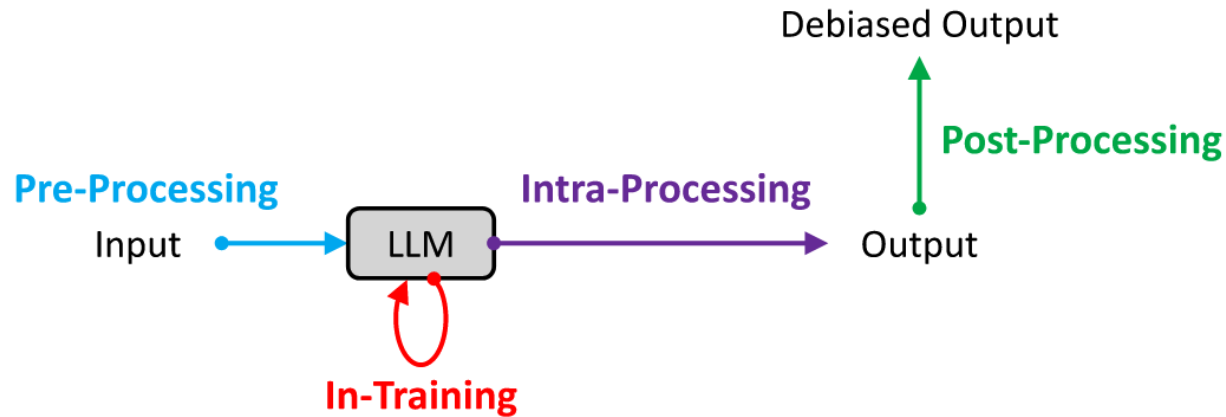
<https://papers.miccai.org/miccai-2025/0819-Paper3129.html>



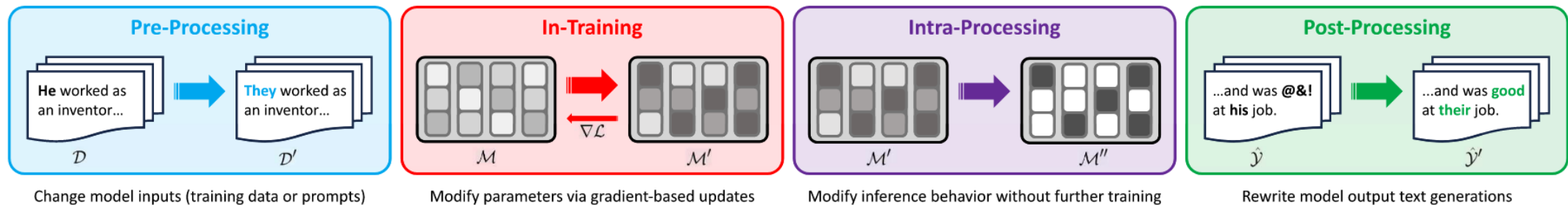
Section 3 – Addressing the Ethical Issues of LLMs

- Fairness and bias
 - Data, evaluation metrics, mitigation methodology
- Accountability
 - Hallucination and uncertainty
- Transparency
 - Approaches and evaluation

How to mitigate bias?



(a)



(b)

Transparency and explainability

Investigate one of the techniques or directions as assigned, read something about it:

1. Attention-based explainability (Sect. 3.1.2)
 - <https://aclanthology.org/P19-3007/>
2. Concept-based explainability (Sect. 3.2.3)
 - <https://aclanthology.org/2022.acl-long.581/> (& for Neuron Activation)
 - <https://doi.ieeecomputersociety.org/10.1109/CVPR52733.2024.01043>
3. ERASER, Faithfulness evaluation of explainability
 - <https://aclanthology.org/2020.acl-main.408/>
4. SHAP (Surrogate Models in Sect. 3.1.1)
 - <https://christophm.github.io/interpretable-ml-book/shap.html>
 - <https://aclanthology.org/2021.hackashop-1.3.pdf>
5. UMAP
 - <https://arxiv.org/abs/1802.03426>
 - <https://pair-code.github.io/understanding-umap/>
 - <https://doi.org/10.1038/s43586-024-00363-x>
6. Other directions: Probing-based (Sect. 3.2.1), Neuron activation (Sect. 3.2.2), mechanistic interpretability (Sect. 3.2.4), Natural Language Explanation (Sect. 3.1.4 and 3.2.2), etc.

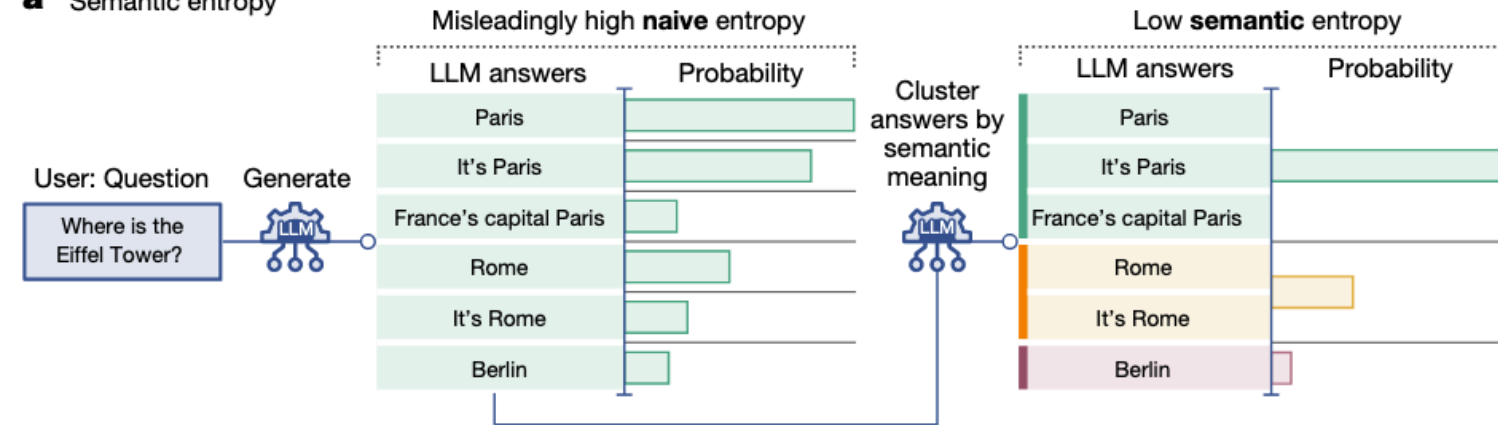
Think about five aspects:

- Intuition of the method
- A use case
- An advantage
- A disadvantage
- Feasibility to implement

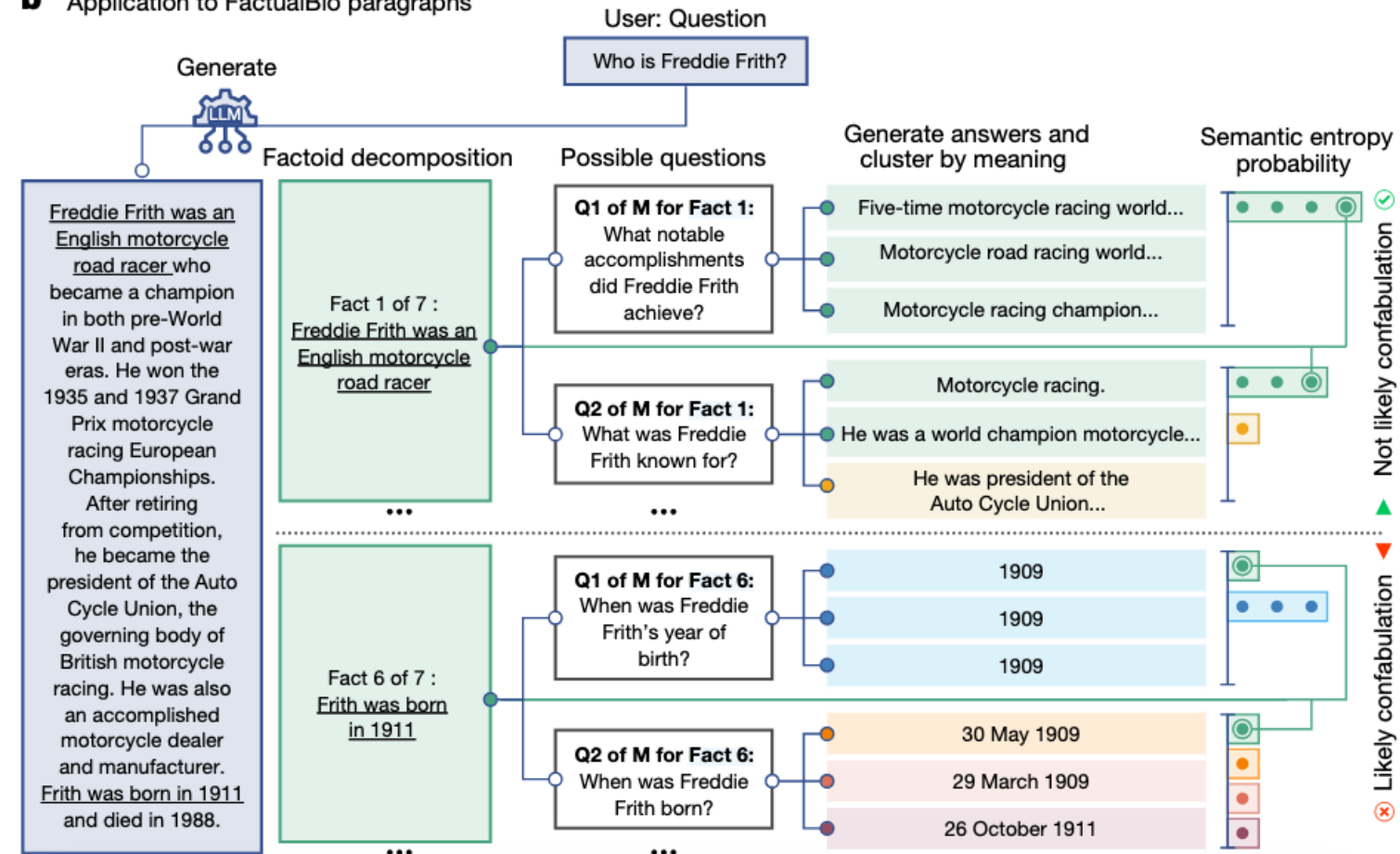
Section numbers are from the survey:
Zhao H, Chen H, Yang F, Liu N, Deng H, Cai H, et al. Explainability for Large Language Models: A Survey. ACM Trans Intell Syst Technol. 2024 Feb 22;15(2):20:1-20:38.
<https://doi.org/10.1145/3639372>

Hallucination & uncertainty

a Semantic entropy



b Application to FactualBio paragraphs



Farquhar, S., Kossen, J., Kuhn, L., & Gal, Y. (2024). Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017), 625–630. <https://doi.org/10.1038/s41586-024-07421-0>

Section 4 - World Models and Autonomous Systems

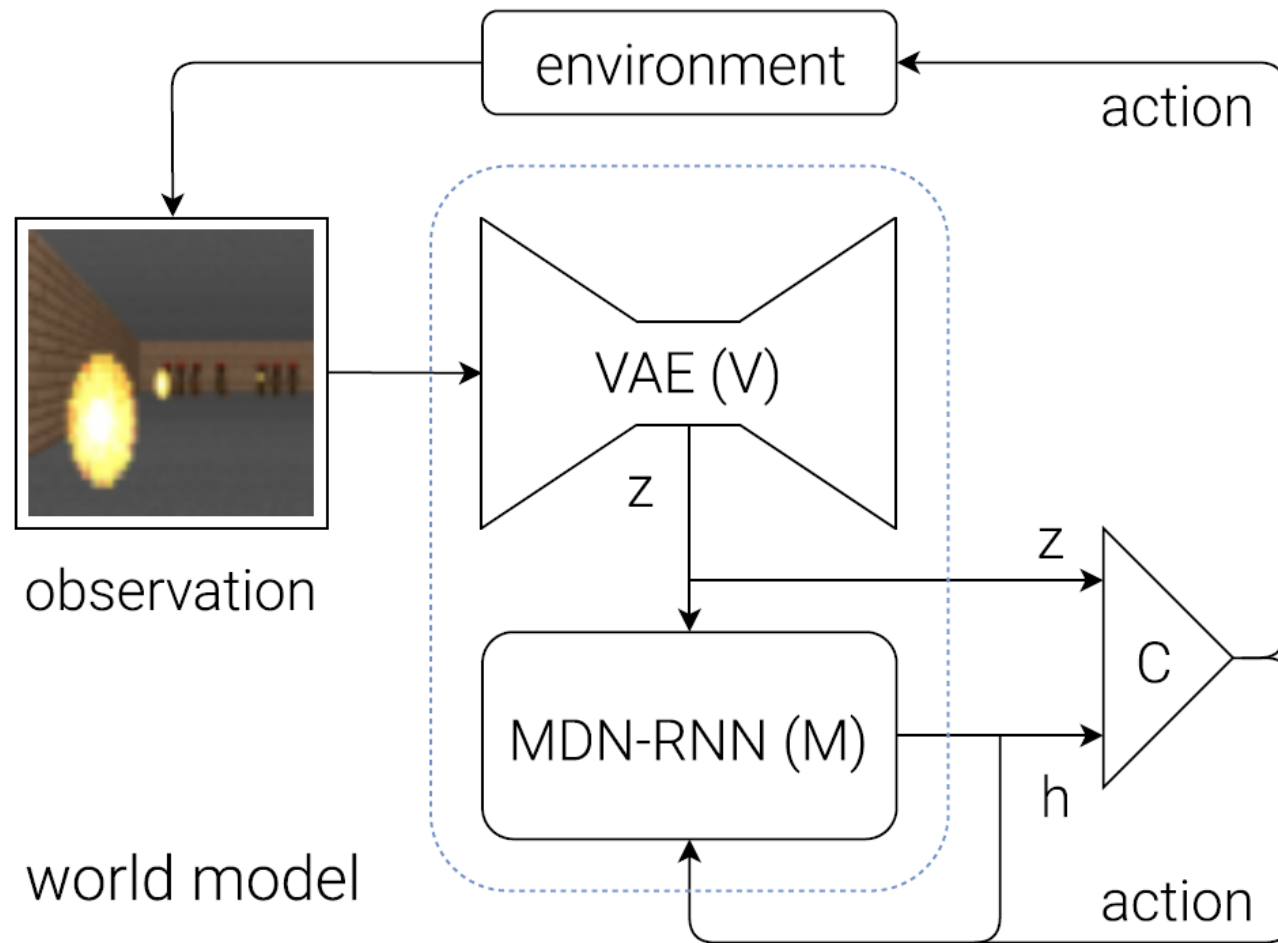


Figure from Ha, D.R., & Schmidhuber, J. (2018). World Models. *ArXiv, abs/1803.10122*.

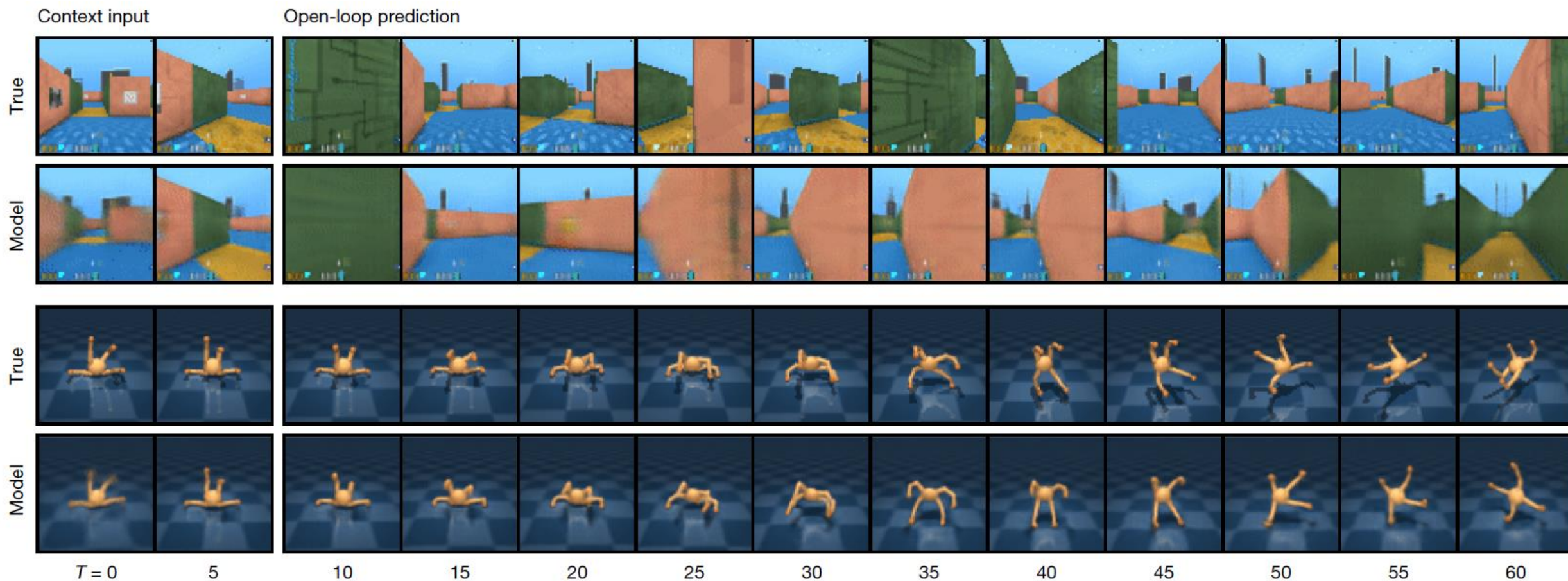


Figure from Hafner, D., Pasukonis, J., Ba, J., & Lillicrap, T. (2025). Mastering diverse control tasks through world models. *Nature*, 640(8059), 647-653. <https://doi.org/10.1038/s41586-025-08744-2>

Section 5 – Domain-specific Applications

- AI for disease progression modelling
 - LLM & KG, explainability, bias
- AI for climate communication
 - Agentic AI, code generation, chatbot, human evaluation
 - LLM for data and knowledge integration
- AI for game development
 - Agentic AI, multimodality
- AI for cultural heritage data management
 - AI and data management, graph database retrieval
- AI for science (hypothesis generation)
 - LLM-as-judge, probabilistic modelling with prompting
- AI for autonomous robotics
 - World model, sensors, safety issues
- AI for software development
 - Agentic AI, human-AI collaboration, etc.

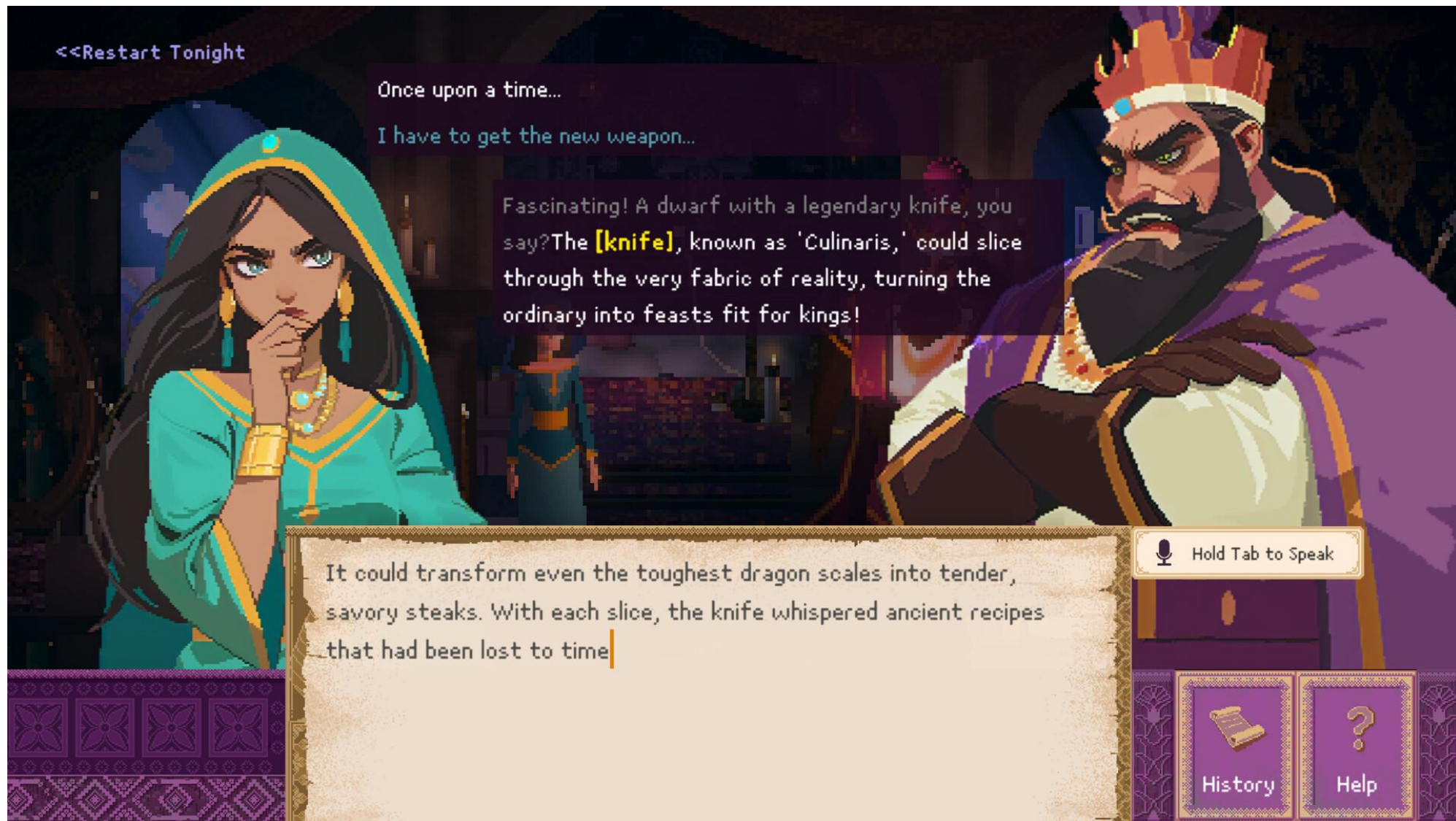
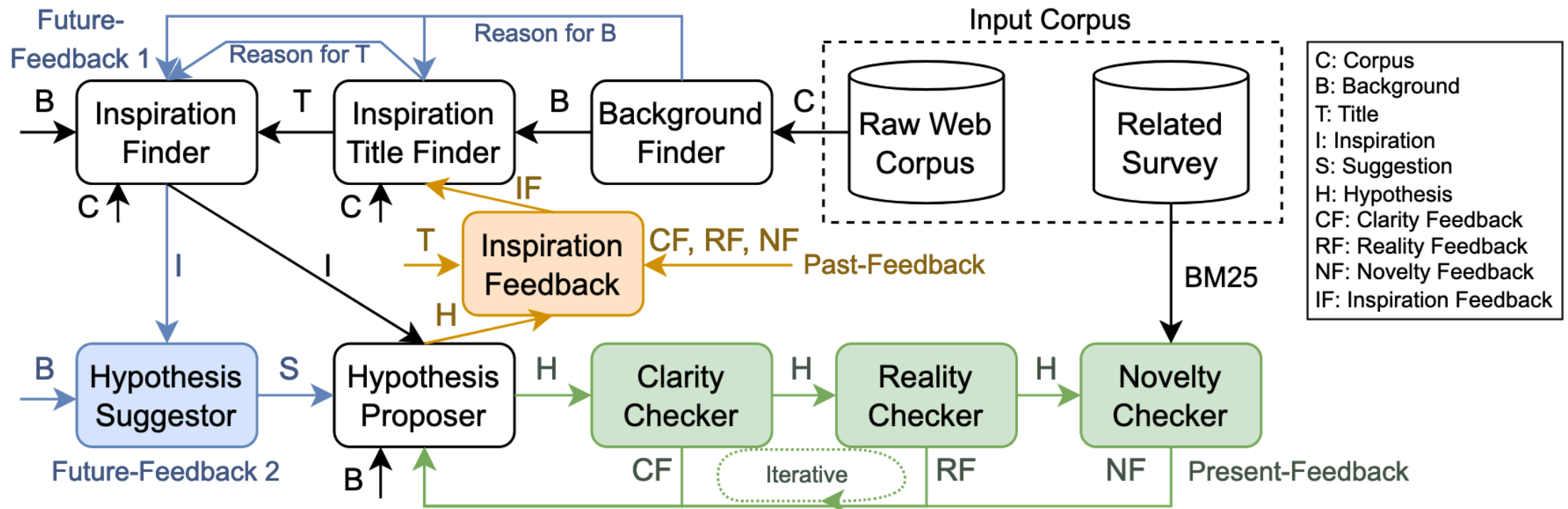
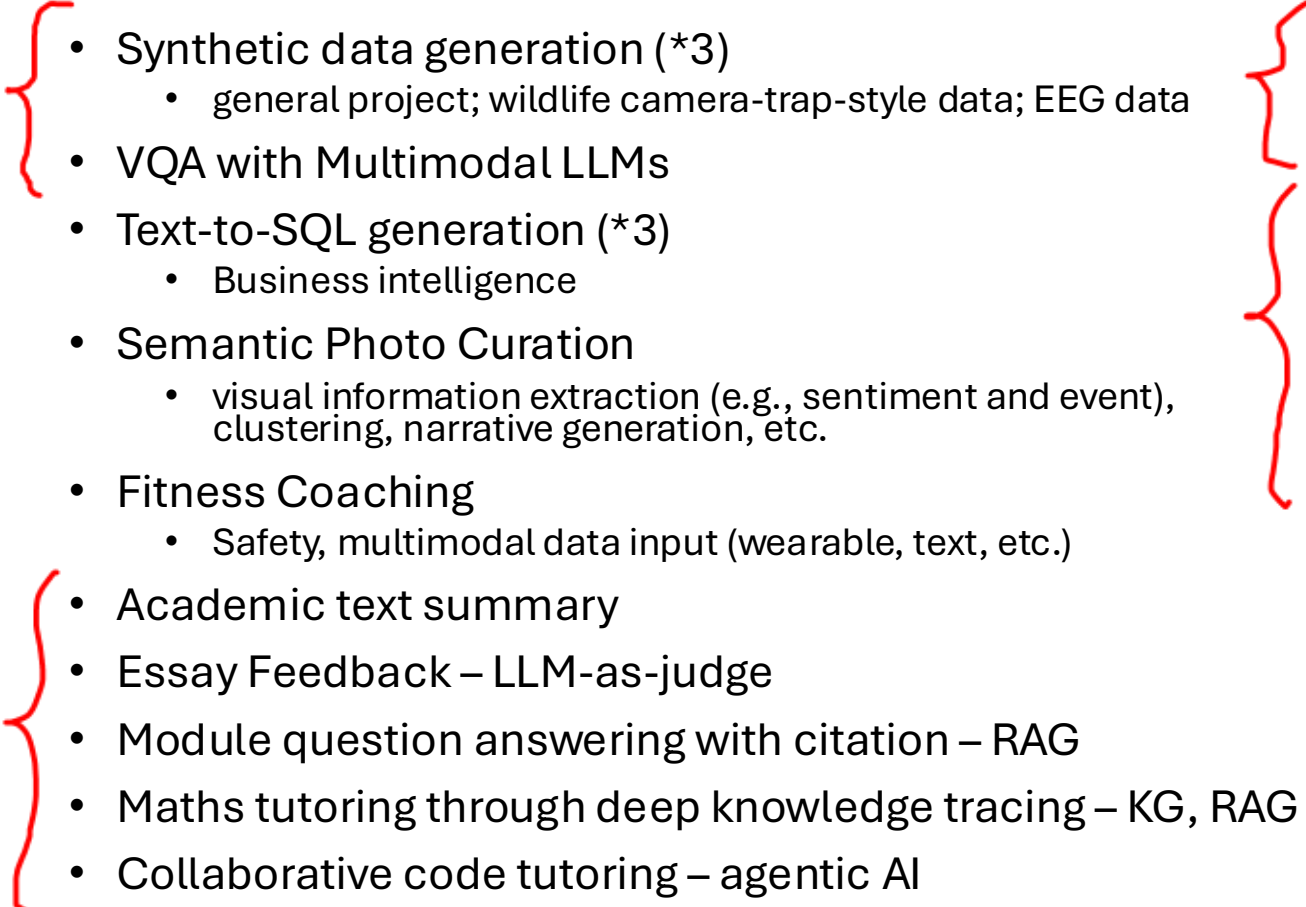


Image from 1001 Nights Demo. Steam. https://store.steampowered.com/app/2782660/1001_Nights_Demo/



Yang, Z., Du, X., Li, J., Zheng, J., Poria, S., & Cambria, E. (2024, August). Large language models for automated open-domain scientific hypotheses discovery. In *Findings of the Association for Computational Linguistics: ACL 2024* (pp. 13545-13565). <https://aclanthology.org/2024.findings-acl.804/>

Your Applications in Coursework 1

- 
- Synthetic data generation (*3)
 - general project; wildlife camera-trap-style data; EEG data
 - VQA with Multimodal LLMs
 - Text-to-SQL generation (*3)
 - Business intelligence
 - Semantic Photo Curation
 - visual information extraction (e.g., sentiment and event), clustering, narrative generation, etc.
 - Fitness Coaching
 - Safety, multimodal data input (wearable, text, etc.)
 - Academic text summary
 - Essay Feedback – LLM-as-judge
 - Module question answering with citation – RAG
 - Maths tutoring through deep knowledge tracing – KG, RAG
 - Collaborative code tutoring – agentic AI
- Radiology report generation
 - EEG data augmentation – Flow Matching
 - Agentic compliance gate for interaction
 - Manufacturing Defect Detection – explainable AI
 - Code Vulnerability Detection
 - Root cause analysis in distributed systems
 - NPC dialogue generation in games – Memory, RAG
 - Visual analysis in games
 - Climate communication – chatbot, RAG, Agentic AI
 - Garment pattern drafting – human-AI co-creation, schema constraints

Some questions for you



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How to learn in the era of GenAI?

- If knowledge is already available on the Web & through GenAI
- In terms of this
 - AI-integrated module
 - Paper reading and discussion
 - Planning a research project in CW1
 - (AI-supported) implementation, e.g., CW2

Accelerate feedback



We would like to ask for your feedback on this module. Your feedback helps us to identify **areas of best practice** as well as **things that can be improved**, so that we can improve your **education and student experience**.

We have set aside 20 minutes to do this. Please:

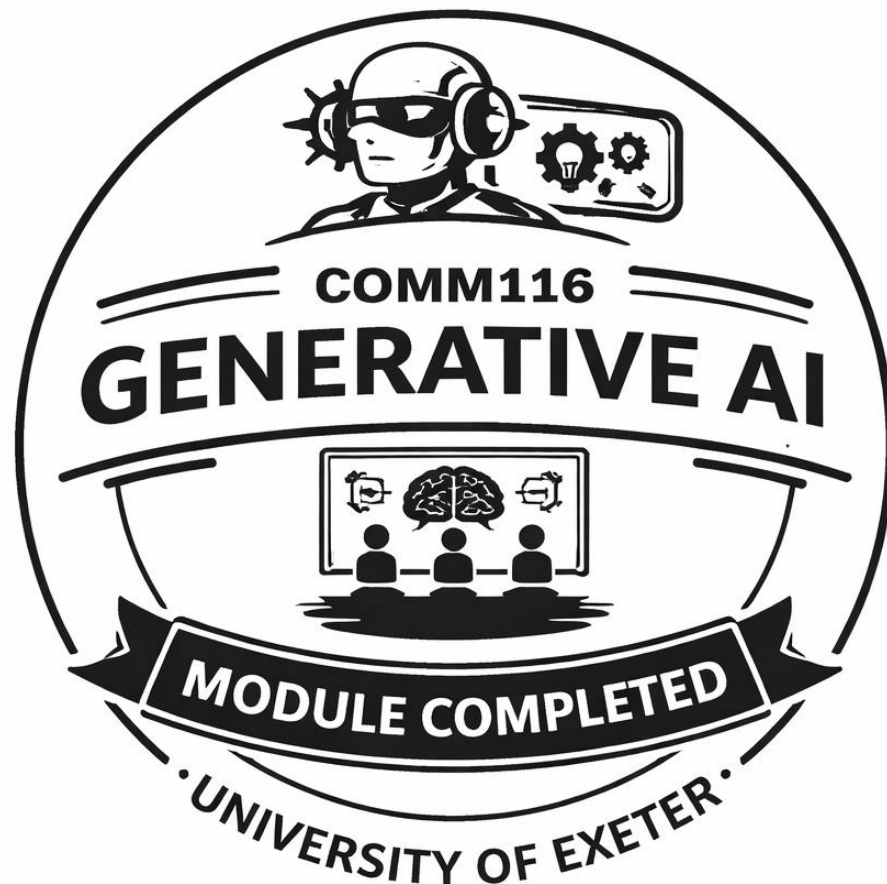
- Spend 10 minutes **discussing your experience** of taking this module with the person or people sitting near to you.
- When you have done this, **complete the Accelerate survey** for the module.

If you would prefer to do this individually you are welcome.

Please ask for help if you need to.



<https://accelerate.exeter.ac.uk/>



Acknowledge to ChatGPT, the full prompt available at <https://chatgpt.com/share/69a58f0e-e9b0-8002-b4b5-1aead6d8f98f>, with the badge above generated on 27 March 2026.